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ABSTRACT

Optical absorption in quantum Bragg mirror detectors (QBMDs) relies on transitions between bound states inside a quantum well and bound or *quasi*-bound states in the continuum in a carefully tailored superlattice. The optimization of QBMDs lattice-matched to InP requires tuning seven specific parameters: the quantum well and barrier thicknesses for the left and right electronic Bragg mirrors (4 parameters), the number of periods in each mirror (2 parameters), and the main quantum well width (1 parameter). We aim to gain insights into the complex QBMD system by mapping transition energies and optimizing heterostructures for enhanced oscillator strength and photocurrent peak amplitude. With all the relevant parameters included in the search space, there are around 50×10^9 possibilities, making it impossible to fully map the search space. Thus, by employing genetic algorithms (GAs) for optimization, heterostructures with optimized oscillator strength and photocurrent amplitude are found, even in vast search spaces. Our investigation demonstrates the feasibility of optimizing QBMD heterostructures using GAs, providing essential design guidelines for improved performance of such devices.

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I. INTRODUCTION

Bound states in the continuum are eigenstates that coexist with the continuum spectrum.¹ This phenomenon is universal to wave physics and has been observed in systems such as acoustics, electronics, optics, and photonics.^{2–5} One possible use of BIC is in infrared detectors. Quantum Bragg mirror detectors (QBMDs)^{6–10} are a class of photodetectors that operate based on optical transitions between a ground state and bound states in the continuum or *quasi*-bound states in the continuum (also called leaky electronic states in the continuum). Compared to traditional quantum well infrared photodetectors (QWIPs),¹¹ QBMDs offer greater flexibility in device design, as the final state of the optical transition is not restricted by the conduction band offset. This enables energy

detection beyond the conduction band offset^{12,13} and simultaneous dual-color detection.¹⁴

All the QBMDs investigated have the following structure: a main quantum well (main QW) (which holds the ground state of the desired optical transition) surrounded by lateral quantum wells (LQWs) on the left and right sides, which compose the electronic Bragg mirrors. This structure gives rise to electronic bound states in the continuum¹⁵ due to constructive/destructive interferences created by the superlattice that composes the quantum Bragg mirrors for electrons. Figure 1(a) depicts one period of the active region of a QBMD structure, showing the electronic probability densities superimposed on the conduction band potential profile. The red line represents the electronic bound states in the continuum. Figure 1(b) shows the experimental photovoltaic photocurrent

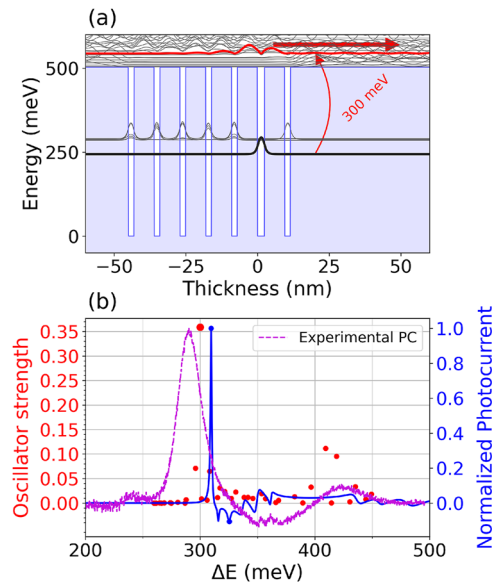


FIG. 1. One period of the active region of the reference QBMD heterostructure. (a) Conduction band profile and electronic probability densities. The red line highlights the first bound state in the continuum (BIC), and the arrow indicates the highest optical transition from the ground state. (b) Normalized experimental (purple line) and simulated (blue line) photovoltaic photocurrent spectra for the structure shown in (a). The red circles correspond to the oscillator strength for optical transitions starting from the ground state.

spectrum¹⁰ (purple line) overlapping the simulated one (blue line). The main photocurrent peak corresponds to the optical transition between the ground state and the first quasi-bound state in the continuum, indicated by the highest oscillator strength (red circles). The design of these structures to obtain a device with specific characteristics requires careful consideration of different parameters, such as dimensions, periodicity and material. The sample with structure and optical characteristics depicted in Fig. 1 will be used as the reference for the reported investigations.

Designing devices with optimized structures often involves extensive calculations or trial-and-error testing. Numerical methods for calculating electronic structure play a crucial role in the development of these photodetectors, as the thicknesses of well and barrier layers significantly impact the detection energy and the creation of quasi-bound states in the continuum in the superlattices. The simulations were performed using the *e-mulate* software^{16,17} developed by our group for lattice-matched InGaAs/InAlAs superlattice structures. The parameters used for the $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}/\text{In}_{0.52}\text{Al}_{0.48}\text{As}$ alloys at 300 K are^{18,19} conduction band offset $V_0 = 503$ meV; electron effective masses $m_{\text{InGaAs}} = 0.043 m_0$ (wells) and $m_{\text{InAlAs}} = 0.076 m_0$ (barriers); and nonparabolicity coefficient $\gamma = 1.0 \times 10^{-18} \text{ m}^{-2}$. The transfer matrix method²⁰ was used to calculate the electronic band structure (wavefunctions and their respective energies) and oscillator strength. The photovoltaic photocurrent spectrum was calculated using the theory of coherent electron emission in quantum well structures.²¹

Genetic algorithms (GAs) represent an optimization technique inspired by the principles of natural selection observed in biological systems.²² This technique can be used in several semiconductor research topics, in which parameter optimization is necessary, such as quantum dots growth,²³ design of resonant-cavity-enhanced photodetectors,²⁴ and high-speed traveling-wave photodetectors.²⁵ GAs operate by generating a group of potential solutions, referred to as a population. These solutions undergo successive iterations and improvements through the application of genetic operators, including mutation and crossover. During each iteration or generation, the fitness of each potential solution within the population is measured, and the fittest solutions are chosen to serve as the foundation for the subsequent generation. By repetitive execution of these processes, GAs efficiently navigate through extensive search spaces, ultimately converging toward structures characterized by enhanced fitness, such as higher oscillator strength and photocurrent amplitude, as observed in our specific case.

The main objectives of this research are (i) mapping transition energies for a range of layer thicknesses within defined boundary limits and (ii) optimizing the heterostructures using genetic algorithms to achieve enhanced oscillator strength and photocurrent amplitude. In order to achieve this first objective, we have calculated the electronic wavefunctions, the transition energies, and the oscillator strengths for various QBMD structures using the reference structure as a starting point. However, as the number of parameters increases, finding high oscillator strength structures by extensive search becomes computationally demanding. For this reason, the mapping process was applied only in a specific range of thicknesses. Within this controlled search space, mapping serves to validate the results later obtained using GA optimization, which is a probabilistic technique.

With the effectiveness of GAs demonstrated in the first objective, we proceed to find structures with a high oscillator strength and photocurrent amplitude in a large search space without the need to map all possible situations. This is desirable especially when the number of possibilities is so large that simulating all of them becomes computationally infeasible.

In the simulations performed in Sec. II (mapping) and Sec. III A (three parameter optimization), the thickness of the main QW, the lateral quantum wells (LQWs) and the lateral quantum barriers (LQBs) are variables, while the quantity of LQW in each side is constant. The LQWs are modified as a group in which all the wells have the same thickness on each simulation. The same restriction applies to the LQBs. In Secs. III B and III C, some restrictions are gradually removed in order to gain insights in the complex task of developing novel QBMDs.

II. MAPPING

The properties of the QBMDs are changed when their design parameters are modified, such as the number of QW on each side, the thicknesses of the LQWs, LQBs, and main QW.

Mapping a range of values enables the creation of a database containing all potential combinations and outcomes of the detectors. This enables the identification of optimized parameter combinations and allows for subsequent comparisons with future optimizations to assess the GA's ability to generate top-ranked solutions. The optimized oscillator strength for a particular transition energy was

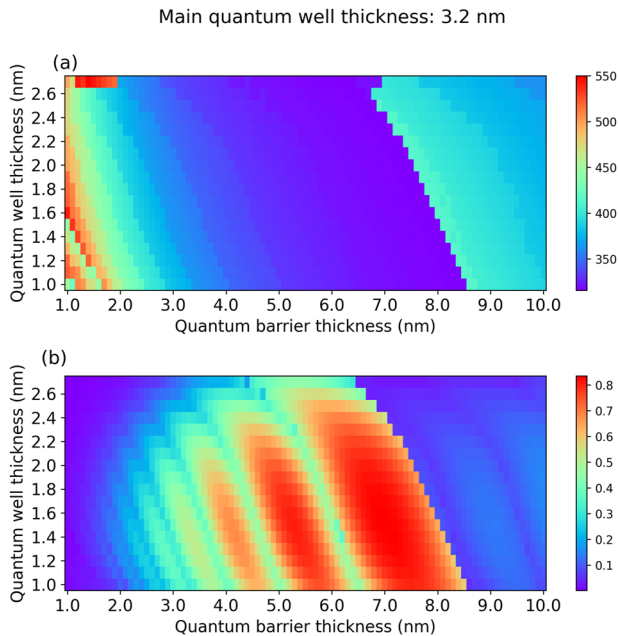


FIG. 2. Transition energy and oscillator strength mapping for samples consisting of one main quantum well thickness fixed at 3.2 nm, lateral quantum well varying between 1.0 and 2.8 nm, and lateral quantum barrier thickness varying between 1.0 and 10.0 nm. (a) Transition energy of the highest oscillator strength and (b) oscillator strength amplitude mapping.

investigated by varying three key parameters of the heterostructure: the thicknesses of the main QW, LQWs, and LQBs. During this mapping procedure, the number of wells on each side (one on the left, five on the right) was kept constant. As expected, this mapping yielded valuable insights into QBMD characteristics and demonstrated the nonlinear correlation between the parameters and the oscillator strength. It additionally identified promising parameters' ranges within the search space. Furthermore, it was observed that specific parameter combinations, even when one parameter was held constant (e.g., main QW thickness), resulted in an oscillator strength four times greater than that of the reference structure. For instance, the simulation map presented in Fig. 2 depicts an example where the main quantum well thickness was 3.2 nm. In this specific map, both the transition energy corresponding to the best oscillator strength [Fig. 2(a)] and the value of the highest oscillator strength [Fig. 2(b)] were mapped as a function of the thicknesses of the LQWs and LQBs. For these color-coded maps, the horizontal axis represents the LQB thicknesses and the vertical axis represents the LQW thicknesses. The corresponding mapping for the photocurrent is available in the [supplementary material](#).

These simulations were conducted on an Intel Core i7-8750H local workstation (2.20 GHz, 6 cores, 12 logical processors, 32 GB RAM). The search space was calculated using a 0.1 nm discretization step and a physical constraint where the main quantum well thickness must be at least 0.5 nm wider than the lateral ones. In particular, the main quantum well thickness was varied in the range of 1.5–10.0 nm, while the lateral quantum barrier and lateral quantum well thicknesses were independently varied in the range of

1.0–10.0 nm. Redundant designs were eliminated by treating the number of wells as an unordered pair to remove “mirror” structures, a method verified exactly by a computerized counter obtained 348 257 possible combinations. However, given that each simulation requires ~30 s, executing all simulations on a single machine was computationally unfeasible. A full sweep would take ~111 days, thus exceeding three months of continuous computation. To mitigate this extensive computational cost, parallelization was employed by leveraging multiple computing cores.^{26,27} Some virtual machines were deployed on Google Cloud Platform (GCP), in addition to the local workstation, in order to facilitate the accomplishment of this task. Two e2-standard-32 virtual machines were used, each configured with 16 physical cores (32 threads) and 128 GB of RAM. This parallel setup reduced the estimated total simulation time from more than 100 days to ~3 days.

III. OPTIMIZATION USING GENETIC ALGORITHMS

As demonstrated in the previous section, mapping all individuals within a limited search space is computationally expensive and time-consuming. Therefore, a genetic algorithm was employed to identify structures with enhanced photocurrent and/or oscillator strength without requiring a full search space simulation.

The photovoltaic QBMD presented in Fig. 1 was used as the initial population for the three series of GA optimizations. Each series progressively increased the number of parameters (variables) by relaxing restrictions to increase simulation complexity. A restriction maintained across all optimizations was that the main QW thickness must be at least 0.5 nm wider than the LQWs, ensuring that the wavefunction of the ground state is localized in the main QW.

- Series 1 (three parameters): The thicknesses of the LQWs, LQBs, and main QW were treated as variables. The number of LQWs on each side of the main QW was fixed at five (left) and one (right). In addition, the thicknesses of the LQWs and LQBs were constrained to be identical on both the left and right sides of the main QW.
- Series 2 (five parameters): The number of LQWs on the left and right sides was also treated as a variable. The restriction that LQW and LQB thicknesses must be equal on both sides of the main QW remained.
- Series 3 (seven parameters): Two additional variables were added by eliminating the equality constraint on the dimensions of LQWs and LQBs on each side.

Table I summarizes the conditions for each optimization series, including the respective search ranges for the GA.

For each series of GA optimizations, three different objective functions were evaluated: optimization of the oscillator strength, photocurrent amplitude, and both oscillator strength and photocurrent amplitude simultaneously. For all GA optimizations, the transition energy was tuned to approximately (300 ± 20) meV, to match the reference structure. While all numerical results are included in the following sections and tables, the figures for single-objective optimizations are provided in the [supplementary material](#) to prioritize space for the simultaneous optimization results.

TABLE I. Parameters and ranges for each GA optimization series. LW: number of lateral quantum wells on left side, left LQW: left lateral quantum well thickness, left LQB: left lateral quantum barrier thickness, main QW: main quantum well thickness, RW: number of lateral quantum wells on right side, right LQW: right lateral quantum well thickness, right LQB: right lateral quantum barrier thickness.

Variables	LW	Left LQW (nm)	Left LQB (nm)	Main QW (nm)	RW	Right LQW (nm)	Right LQB (nm)
Three	5 ^a	[1.0–10.0]	[1.0–10.0]	[1.0–10.0]	1 ^a	... ^b	... ^b
Five	[1–7]	[1.0–10.0]	[1.0–10.0]	[1.0–10.0]	[1–7]	... ^b	... ^b
Seven	[1–7]	[1.0–10.0]	[1.0–10.0]	[1.0–10.0]	[1–7]	[1.0–10.0]	[1.0–10.0]

^aFixed parameter in the optimization.

^bThicknesses equal to the left side.

A. Series 1: Three-parameter optimization

The objective function aimed the enhancement of the oscillator strength beyond that of the reference structure value of 0.36. The best optimized individual is composed of main QW, LQW, and LQB with thicknesses of 3.2, 1.5, and 7.0 nm, respectively. This heterostructure presents an oscillator strength of 0.83 for an optical transition at 318.8 meV, representing a 132.6% increase over that of the reference structure. As expected, this optimized structure exhibits a bound-to-quasi-bound transition, since the wavefunction overlap between the initial and final electronic states is larger than that of the case of bound to continuum transitions.

Comparing the GA results with the complete search space mapping helps elucidate the advantages of using genetic algorithms. Table II presents the best individual from each of ten optimizations, performed under identical initial conditions, along with their respective rank positions within all simulated possibilities. These optimizations consistently provide a solution ranked among the top 20 of the entire search space. Furthermore, across these ten GA runs, the best solution identified successfully matched the global optimum.

Moreover, while the GA does not guarantee finding the global optimum, it identifies solutions that are at least as good as the

reference, never converging to a worse outcome. This highlights its robustness in optimization tasks.

It is important to note that the GA was specifically configured to optimize the oscillator strength exclusively, without considering the photocurrent amplitude. As a result, while the optimized solution showed a high oscillator strength (e.g., around 0.83), the photocurrent amplitude decreased to 0.77, a value 22.8% lower than that of the reference structure.

In the second optimization, the photocurrent amplitude was set as the objective function for maximization. The resulting optimized structure features main QW, LQW, and LQB thicknesses of 3.0, 1.8, and 10.0 nm, respectively. This configuration yielded a photocurrent of 1.53 representing a 53% gain compared to the reference sample's value of 1.00. The oscillator strength is 65% higher than that of the reference structure, but it remains lower than the value achieved by the OS optimization. This outcome might initially appear counterintuitive, but a high photocurrent requires not only efficient absorption (indicated by oscillator strength) but also effective carrier extraction. However, a high oscillator strength typically implies that confined states, which can, in turn, lead to reduced carrier extraction efficiency.

Finally, the third optimization combined the objectives of the previous two, simultaneously seeking to maximize both oscillator strength and photocurrent amplitude.

For this dual optimization, the fitness contributions from each objective were combined. The oscillator strength was adjusted by multiplying it by a fixed factor [Eq. (1)], while the photocurrent amplitude was proportionally scaled by an oscillator strength value to ensure both metrics were on the same order of magnitude and directly comparable [Eq. (2)]. These two scaled fitness values were then summed [Eq. (3)], providing similar weight to both parameters during the optimization process. The specific proportional values used in these equations were empirically determined to facilitate the identification of solutions that simultaneously yield favorable results for both parameters, as follows:

$$OS_{fitness} = OscillatorStrength * 20, \quad (1)$$

$$PC_{fitness} = \log_{10}(|PC|) + 30, \quad (2)$$

$$Structure_{fitness} = OS_{fitness} + PC_{fitness}. \quad (3)$$

TABLE II. Best individual and classifications for 10 optimizations. LQW: lateral quantum well thickness, LQB: lateral quantum barrier thickness, main QW: main quantum well thickness, OS: oscillator strength, TE: transition energy for best oscillator strength.

Ranking	LQW (nm)	LQB (nm)	Main QW (nm)	OS	TE (meV)
1	1.5	7.0	3.2	0.835	318.75
1	1.5	7.0	3.2	0.835	318.75
7	1.4	7.2	3.2	0.832	318.39
2	1.5	7.1	3.2	0.834	318.41
1	1.5	7.0	3.2	0.835	318.75
16	1.3	7.1	3.2	0.827	319.06
2	1.5	7.1	3.2	0.834	318.41
1	1.5	7.0	3.2	0.835	318.75
1	1.5	7.0	3.2	0.835	318.75
2	1.5	7.1	3.2	0.834	318.41

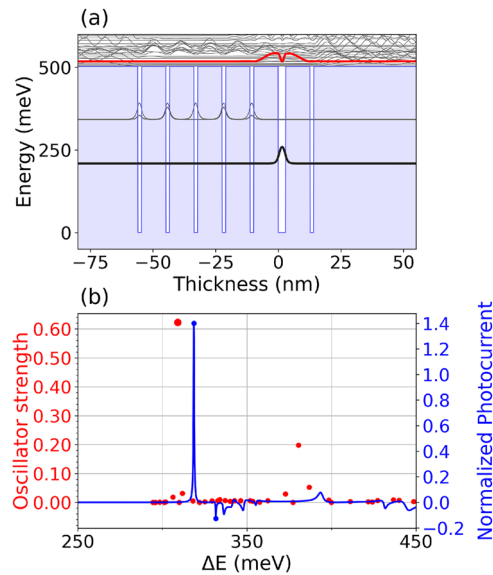


FIG. 3. Heterostructure obtained from the three-parameter optimization of oscillator strength and photocurrent amplitude simultaneously. (a) Conduction band profile and electronic probability density. (b) Photovoltaic photocurrent spectrum (blue line) and oscillator strength (red circles) for transitions starting from the ground state.

Figure 3(a) illustrates the optimized structure derived from simultaneously optimizing oscillator strength and photocurrent amplitude. This configuration features the main QW, LQW, and LQB with thicknesses of 3.0, 1.5, and 9.7 nm, respectively. The optimization enhanced the oscillator strength from 0.36, for the reference, to 0.62, with a 73.4% gain. On the other hand, the photocurrent amplitude improved from 1.00 to 1.40 reaching a 40%. These optimized dimensions are remarkably similar to those of the structure optimized solely for oscillator strength, differing primarily in the barrier, which is ~ 2.0 nm larger. This observation highlights the critical importance of including both oscillator strength and photocurrent amplitude in the optimization process. The summary of the results for all three optimization series is presented in Table III.

B. Series 2: Five-parameter optimization

This section details a five-parameter optimization, building upon the three variables from the previous series by varying the number of lateral quantum wells (LQWs) on each side of the main QW. This expansion allows for solutions from one to seven LQWs on each side.

It is important to note that this expanded search space is significantly larger, rendering total simulation nearly impossible. Instead of the $\sim 330\,000$ possible simulations encountered previously, the number has now increased to almost 9×10^6 .

Similar to the previous optimization series (Subsection III A), three distinct optimizations were conducted, each targeting specific objectives: optimizing only the oscillator strength, only the photocurrent amplitude, and both simultaneously.

The search space for QBMD simulations exhibits symmetry according to the number of lateral wells on the left and right sides. That means, a structure with five wells on the left and two on the right presents the same result as a structure with two wells on the left and five on the right, assuming constant dimensions for the barriers, lateral wells, and main well.

Figure 4 presents the heterostructure obtained from the five-parameter optimization of oscillator strength and photocurrent amplitude simultaneously. Figure 4(a) shows the conduction band profile and electronic probability densities. In Fig. 4(b), the photovoltaic photocurrent spectrum (blue line) and oscillator strength (red circles) are plotted for transitions starting from the ground state.

The results for the best individual found in each optimization are displayed in Table IV (oscillator strength optimization, photocurrent amplitude optimization, and simultaneous optimization), and in Fig. 4, the structure and results for simultaneous optimization are shown. For oscillator strength optimization, its value increased from 0.36, for the reference, to 0.93, representing a 158.8% gain. For photocurrent optimization, the value spanned from 1.00 (reference) to 39.40 resulting in a 3840% gain. The simultaneous optimization yielded an oscillator strength ranging from 0.36 to 0.59 achieving up to 63.4% gain, and a photocurrent from 1.00 (reference) to 135 resulting in up to 13 400% gain. The substantial photocurrent gain observed suggests that the BIC achieves an improved balance between the overlap with the ground state and the electron escape probability. This simultaneous optimization is evidenced by the narrow linewidth of the photocurrent peak presented in Fig. 4(b).

TABLE III. Three optimization parameters, best results for each optimization compared with the reference structure. OS: oscillator strength, TE: transition energy for best oscillator strength, PC: photocurrent amplitude, PC E: energy of the photocurrent peak, OS gain: oscillator strength gain, and PC gain: photocurrent amplitude gain.

Objective function	Structure			Oscillator strength		Photocurrent		Gain	
	LQW (nm)	LQB (nm)	Main QW (nm)	OS	TE (meV)	PC	PC E (meV)	OS gain (%)	PC gain (%)
Reference (Fig. 1)	2.0	7.0	2.5	0.36	299.9	1.00	309.6	...	...
Oscillator strength	1.5	7.0	3.2	0.83	318.8	0.77	335.2	132.6	-22.8
Photocurrent	1.8	10.0	3.0	0.59	307.8	1.53	316.8	65.1	53
Oscillator strength and photocurrent (Fig. 3)	1.5	9.7	3.0	0.62	309.0	1.40	318.6	73.4	40

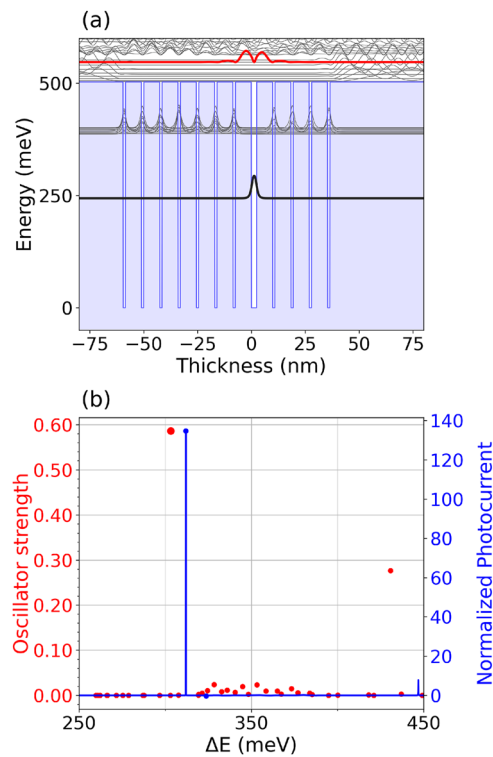


FIG. 4. Heterostructure obtained from the five-parameter optimization of oscillator strength and photocurrent amplitude simultaneously. (a) Conduction band profile and electronic probability density. (b) Photovoltaic photocurrent spectrum (blue line) and oscillator strength (red circles) for transitions starting from the ground state.

Table IV summarizes the values of all three optimizations, including their improvements compared to the reference structure. It is important to note that some optimizations yielded structures with seven lateral wells on one side, which represents the upper limit imposed within the genetic algorithm’s search space. This limit was arbitrarily set for growth purposes only.

C. Series 3: Seven-parameter optimization

To further test the genetic algorithm’s ability to find optimized solutions in larger search spaces, optimizations with seven

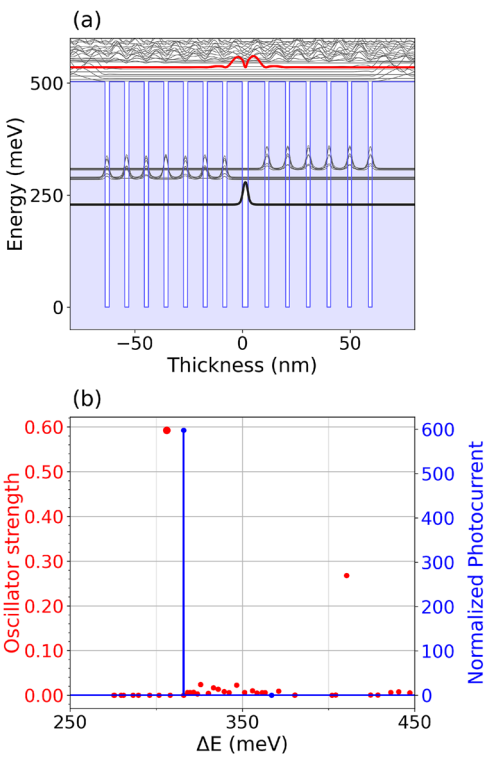


FIG. 5. Heterostructure with the highest oscillator strength and photocurrent amplitude obtained from the seven-parameter optimization. (a) Conduction band profile and electronic probability density. (b) The normalized photovoltaic photocurrent spectrum (blue line) and oscillator strength (red circles).

parameters were performed. This series expanded upon the previous five-parameter case, by allowing the dimensions of the left-side wells and barriers to differ from those on the right, thus introducing an additional asymmetry into the design. The seven search parameters now include the number of left LQWs, thickness of left LQW, thickness of left LQB, thickness of the main QW, number of right LQWs, thickness of right LQW, and thickness of right LQB. With this increased complexity, the search space has expanded to encompass 50×10^9 possibilities, which would require over 47 000 years of serial computation.

TABLE IV. Five optimization parameters, best results for each optimization compared with the reference structure. LW: number of lateral quantum wells on left side and RW: number of lateral quantum wells on right side.

Objective function	Structure					Oscillator strength		Photocurrent		Gain	
	LW	LQW (nm)	LQB (nm)	Main QW (nm)	RW	OS	TE (meV)	PC	PC E (meV)	OS gain (%)	PC gain (%)
Reference (Fig. 1)	5	2.0	7.0	2.5	1	0.36	299.9	1.00	309.6	...	...
Oscillator strength	5	1.9	6.5	3.2	2	0.93	319.4	8.8	336.3	158.8	780
Photocurrent	6	2.0	7.0	2.5	4	0.45	298.5	39.40	308.2	25.0	3840
Oscillator strength and photocurrent	7	1.1	7.4	2.5	4	0.59	303.2	135	312.0	63.4	13 400

TABLE V. Seven optimization parameters and best results for each optimization compared with the reference structure. Left LQW: left lateral quantum well thickness, left LQB: left lateral quantum barrier thickness, right LQW: right lateral quantum well thickness, and right LQB: right lateral quantum barrier thickness.

Objective function	Structure							Oscillator strength		Photocurrent		Gain	
	LW	Left LQW	Left LQB	Main QW	RW	Right LQW	Right LQB	OS	TE (meV)	PC	PC E (meV)	OS gain (%)	PC gain (%)
		(nm)	(nm)	(nm)		(nm)	(nm)						
Reference (Fig. 1)	5	2.0	7.0	2.5	1	2.0	7.0	0.36	299.9	1.00	309.6	...	...
Oscillator strength	7	2.2	5.9	3.2	3	1.6	7.0	0.94	319.5	0.26	336.5	162.1	−73.8
Photocurrent	7	1.8	8.8	2.7	3	1.4	9.4	0.47	299.7	63.1	306.8	30.7	6210
Oscillator strength and photocurrent (Fig. 5)	7	2.0	7.1	2.7	6	1.8	7.8	0.59	306.1	597	315.9	65.0	59 600

Maintaining the sequence of previous optimizations, three distinct optimizations were carried out to enhance oscillator strength, improve photocurrent, and optimize both parameters simultaneously. The outcome for the most favorable individuals identified in each scenario is as follows: For oscillator strength optimization, its value increased from 0.36 to 0.94, representing a 162.1% gain. For photocurrent optimization, the value increased from 1.00 to 63.1 showing a substantial 6210% gain. The simultaneous optimization (Fig. 5) yielded an oscillator strength increase from 0.36 to 0.59 (65.0% gain) and a photocurrent increase from 1.00 to 597 (59 600% gain).

The heterostructure obtained from the five-parameter optimization of oscillator strength and photocurrent amplitude simultaneously. Figure 5(a) shows the conduction band profile and electronic probability density. Figure 5(b) shows the photo-voltaic photocurrent spectrum (blue line) and oscillator strength (red circles) for transitions starting from the ground state.

Table V provides a comprehensive summary of results of these three optimizations, including their improvements compared to the reference sample.

IV. CONCLUSIONS

This work demonstrates the significant advantages of computer simulations and parameter optimization in the development of novel quantum Bragg mirror detectors (QBMDs). Two distinct simulation approaches were employed: a detailed parameter mapping of QBMD behavior based on their dimensions and the utilization of genetic algorithms (GAs) to identify optimized QBMDs structures without requiring an exhaustive search space mapping.

The detailed mapping was conducted in a restricted three-parameter space due to significant computational time limitations. This map served as a benchmark and comparisons revealed that the GA consistently identified solutions ranked among the best within the mapped structures. The GA pursued three distinct optimization objectives: maximize oscillator strength, maximize photocurrent amplitude, and optimize both simultaneously for a target 300 meV optical transition. The investigation extended these GA simulations beyond the initial three-parameter mapped space to include five- and seven-parameter optimizations. In all series, the GA provided heterostructures with superior oscillator strength and

photocurrent amplitude compared to the reference one. The predicted photocurrent amplitude exhibited substantial gains: 40% for the three-parameter optimization, 13 400% for the five-parameter, and 59 600% for the seven-parameter optimization, when oscillator strength and photocurrent amplitude were optimized simultaneously. The photocurrent gains obtained from the GA optimizations should be interpreted as an upper limit for a real device, as the model does not include electron scattering, interface roughness, or other mechanisms, which inevitably limit the experimental performance.

The results presented in this report demonstrate the GA’s capability to navigate vast search spaces and identify optimized QBMD heterostructures, in scenarios where comprehensive mapping is computationally infeasible. The integrated use of GAs, computer simulations, and mapping trends offers a promising approach for tailoring QBMD heterostructures for specific applications. The designed photodetector structures shall be grown to validate the theoretical work.

SUPPLEMENTARY MATERIAL

The supplementary material associated with this article includes additional simulation results to support the main study. It contains the mapping for the photocurrent peak energy and photocurrent peak intensity, and the optimization figures (structural diagrams and performance plots) for single-objective functions targeting only oscillator strength or only photocurrent intensity.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jose E. Ruiz: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Resources (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Pedro H. Pereira:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Guilherme M. Torelly:** Software (equal); Writing – review & editing (equal). **Patricia L. Souza:** Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal). **Mauricio P. Pires:** Funding acquisition (equal); Investigation (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **Germano M. Penello:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The code for the simulation software is freely available at <https://github.com/LabSem-PUC-Rio/e-mulate>.

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